STATISTICS NOTES

Binomial Distribution • Normal Distribution • Central Limit Theorem

Study guide — definitions, conditions, formulas, worked example and key comparisons.

1. BINOMIAL DISTRIBUTION

Definition

A binomial distribution is a probability distribution that describes the probability of obtaining a specific number of successes in a fixed number of independent trials, where each trial has only two possible outcomes (success or failure), and the probability of success remains constant in every trial.

In simple English: It helps us calculate the likelihood of something happening when an experiment is repeated a fixed number of times.

Mathematical notation

X ~ B(n, p)

X = random variable (number of successes); **B** = binomial distribution; **n** = number of trials; **p** = probability of success; **q** = probability of failure.

q = 1 – p **and** **p + q = 1**

Conditions for a binomial distribution

1. **Fixed number of trials:** The number of attempts is known in advance.
2. **Two possible outcomes:** Each trial has only two outcomes, called success and failure.
3. **Independent trials:** The outcome of one trial does not affect another.
4. **Constant probability:** The probability of success is the same in every trial.

Real-life examples

1. Tossing a coin 10 times and finding the probability of exactly 6 heads.
2. Finding the probability that exactly 15 out of 20 students pass an examination, assuming the same pass probability and independent outcomes.
3. Finding the probability of exactly 4 defective bulbs among 50 bulbs, assuming a constant defect probability and independent outcomes.

Binomial formulas

Mean: $\mu = np$

Variance: $\sigma^2 = npq$

Standard deviation: $\sigma = \sqrt{(npq)}$

Probability of exactly x successes: $P(X = x) = C(n,x)\ p\blacksquare\ q\blacksquare\blacksquare\blacksquare\blacksquare$

Here, C(n,x) is the number of ways to choose x successes from n trials.

2. NORMAL DISTRIBUTION

Definition

A normal distribution is a continuous probability distribution that is bell-shaped and symmetrical around its mean. It describes how data values are distributed around the average.

In simple English: Most values are concentrated around the centre, while fewer values appear towards the two ends.

Characteristics

1. It is continuous.
2. It is bell-shaped and symmetrical around the mean.
3. The mean, median and mode are equal.
4. The total area under the curve is 1 (100%).
5. The two tails extend indefinitely and approach, but do not touch, the horizontal axis.

Mathematical notation

$$\mathbf{X \sim N(\mu, \sigma^2)}$$

X = random variable; **N** = normal distribution; μ = mean; σ = standard deviation; σ^2 = variance.

Mean (μ)

The mean determines the location or centre of the normal distribution curve. It is the average value of the data.

$$\textbf{Mean} = \textbf{(sum of all observations)} \div \textbf{(number of observations)}$$

Example: For scores 40, 50, 60, 70 and 80:
Mean = (40 + 50 + 60 + 70 + 80) ÷ 5 = 300 ÷ 5 = **60**.

Standard deviation (σ) and variance (σ^2)

Standard deviation measures the spread of data around the mean. A small standard deviation means values are closer to the mean; a large standard deviation means values are more spread out.

Variance measures spread using squared differences from the mean. Variance is the square of standard deviation: $\sigma^2 = (\sigma)^2$.

Effects of changing the mean and standard deviation

Changing the mean: Increasing the mean moves the curve to the right. Decreasing the mean moves it to the left. If standard deviation stays constant, the spread does not change.

Changing standard deviation: Increasing standard deviation makes the curve wider and flatter. Decreasing it makes the curve narrower and taller. If the mean stays constant, the centre does not change.

3. BINOMIAL-TO-NORMAL APPROXIMATION

A binomial distribution can sometimes be approximated by a normal distribution when the number of trials is sufficiently large and both expected counts of successes and failures are sufficiently large.

Common classroom conditions

$$\mathbf{np \geq 5} \quad \textbf{and} \quad \mathbf{nq \geq 5}$$

Some textbooks use the stricter guideline $np \geq 10$ and $nq \geq 10$. Follow the convention used by your lecturer or course notes.

If the approximation is appropriate:

$$\mathbf{X \approx N(np, npq)}$$

The normal distribution uses mean np and variance npq. Its standard deviation is $\sqrt{(npq)}$.

4. WORKED EXAMPLE: 50 COIN TOSSES

Question: A fair coin is tossed independently 50 times. Let X be the total number of heads obtained.
1. State the distribution of X.
2. Calculate its mean, variance and standard deviation.
3. Show that the conditions for normal approximation are satisfied.

Step 1: Identify the information

Number of trials: n = 50
Probability of a head: p = 0.5
Probability of a tail: q = 1 – 0.5 = 0.5

Step 2: State the distribution

$X \sim B(n,p) = B(50, 0.5)$

Step 3: Calculate the mean

$\mu = np = 50 \times 0.5 = 25$

Step 4: Calculate the variance

$\sigma^2 = npq = 50 \times 0.5 \times 0.5 = 12.5$

Step 5: Calculate the standard deviation

$\sigma = \sqrt{12.5} = 3.54$ (approximately)

Step 6: Check the normal approximation conditions

First condition: $np = 50 \times 0.5 = 25 \geq 5$. Satisfied.
Second condition: $nq = 50 \times 0.5 = 25 \geq 5$. Satisfied.

Conclusion: Both conditions are satisfied, so a normal approximation is appropriate.
$X \approx N(25, 12.5)$ (mean = 25; variance = 12.5).

5. CENTRAL LIMIT THEOREM (CLT)

Definition

The Central Limit Theorem states that when sufficiently large random samples are taken from a population, the distribution of their sample means will be approximately normal, regardless of the shape of the original population, provided the observations are independent and the population has a finite variance.

In simple English: Take many random samples of the same size, calculate the mean of each sample, and make a graph of those means. As the sample size increases, the distribution of the sample means tends to become approximately bell-shaped.

Example

Imagine a university has 1,000 students. Randomly select 30 students and calculate their average age. Repeat this many times, each time selecting another sample of 30 and calculating its average age. The many sample means form a distribution that tends towards a normal shape.

Mathematical formula

$\bar{X} \approx N(\mu, \sigma^2/n)$

$\bar{X}$ = sample mean; μ = population mean; σ^2 = population variance; **n** = sample size.

Important CLT formulas

Mean of the sample means: $\mu_{\bar{x}} = \mu$

Variance of the sample means: $\sigma^2_{\bar{x}} = \sigma^2/n$

Standard error: $SE = \sigma/\sqrt{n}$

Conditions and guidance

1. Samples should be randomly selected.
2. Observations should be independent. For sampling without replacement, the sampling fraction should generally be small (a common guideline is no more than 10% of the population).
3. The population should have finite variance.
4. The sample size should be sufficiently large. A sample size of 30 is often used as a classroom guideline, but the required size depends on the population's shape and other conditions.

6. SUMMARY OF IMPORTANT FORMULAS

Concept	Formula
Binomial mean	$\mu = np$
Binomial variance	$\sigma^2 = npq$
Binomial standard deviation	$\sigma = \sqrt{npq}$
Failure probability	$q = 1 - p$
Binomial notation	$X \sim B(n,p)$
Normal notation	$X \sim N(\mu,\sigma^2)$
Normal approximation	$X \approx N(np,npq)$
CLT sample mean	$\mu_{\bar{x}} = \mu$
CLT sample variance	$\sigma^2_{\bar{x}} = \sigma^2/n$
Standard error	$SE = \sigma/\sqrt{n}$

7. IMPORTANT DIFFERENCES

Binomial distribution vs Normal distribution

Binomial	Normal
Discrete distribution	Continuous distribution
Counts successes in a fixed number of trials	Describes continuous values around a mean
Each trial has two possible outcomes	Can take many values over a continuous range
Uses n and p	Uses μ and σ^2
Written B(n,p)	Written $N(\mu,\sigma^2)$

Normal distribution vs Central Limit Theorem

Normal distribution	Central Limit Theorem
Describes a normal-shaped probability distribution.	Explains why sample means tend towards a normal distribution.
Can describe individual observations or a random variable.	Describes the distribution of sample means.
Uses mean and standard deviation.	Uses population mean and standard error.

Revision reminder: Mean tells you the centre; standard deviation tells you the spread; variance is the square of standard deviation. For a binomial distribution, remember $\mu = np$, $\sigma^2 = npq$, and $\sigma = \sqrt{npq}$.